

6S SENSORLESS BLDC ESC

Design Documentation

1. Objective:

The primary objective of this project is to design and develop a reliable 6S Sensor less Brushless DC (BLDC) Electronic Speed Controller (ESC) for drone propulsion systems. The ESC is intended to efficiently control the speed and operation of a three-phase BLDC motor using sensor less Back-EMF-based commutation, eliminating the need for Hall-effect sensors. The project aims to integrate a high-performance STM32G071 microcontroller, FD6288Q gate driver, and a three-phase MOSFET inverter to achieve efficient motor control, real-time current monitoring, and robust protection features.

The project also focuses on implementing accurate Back-EMF detection, current sensing, overcurrent protection, and stable power management through dedicated 12V and 3.3V supply rails. Additionally, the design emphasizes reliability, efficiency, and compactness by incorporating proper gate-drive circuitry, bootstrap networks, transient protection, and optimized PCB design practices.

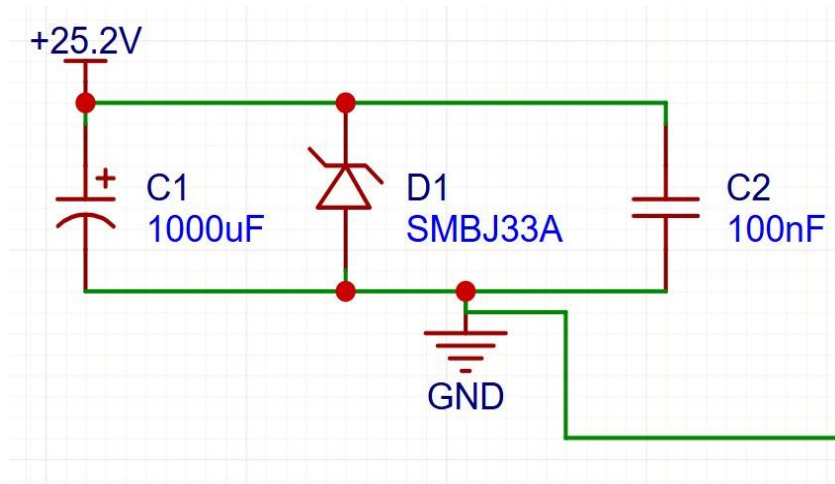
The development follows a phased approach, where the complete ESC schematic is first designed, simulated, and verified in EasyEDA to ensure correct functionality of all subsystems. Once the schematic design is thoroughly reviewed and validated, the project will proceed to PCB design, hardware fabrication, assembly, and experimental testing. The final goal is to develop a lightweight, high-performance ESC suitable for drone applications, providing efficient motor operation, enhanced safety, and reliable performance under real flight conditions.

2. Power input protection circuit

Components:

- Battery connector
- TVS Diode — SMBJ33A
- Bulk electrolytic capacitors

Schematic circuit:



Explanation:

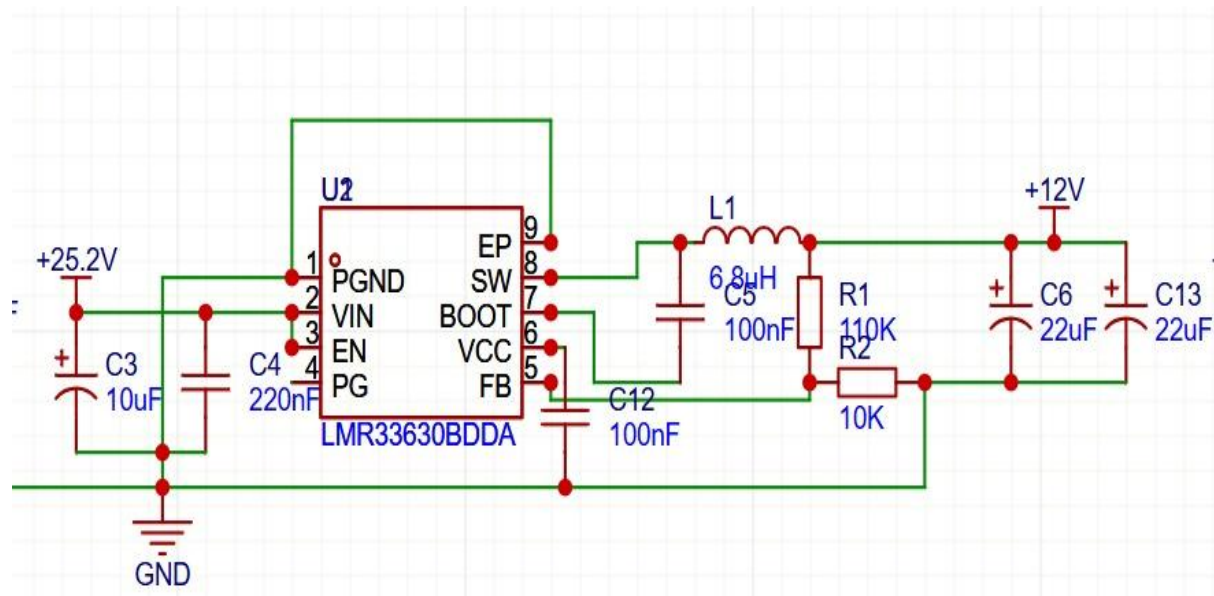
The power input protection circuit is designed to protect the ESC from voltage transients, current surges, and electrical noise generated by the 6S LiPo battery (18V–25.2V) and the switching action of the three-phase inverter. The SMBJ33A TVS diode clamps harmful voltage transients, protecting the STM32 microcontroller, gate driver, and MOSFETs from overvoltage damage. A 1000 μ F, 50V electrolytic capacitor provides energy during sudden load changes and reduces voltage dips, while a 100nF, 50V ceramic capacitor filters high-frequency switching noise and electromagnetic interference (EMI). Together, these components ensure a stable power supply, enhance system reliability, and improve the overall safety and performance of the ESC during drone operation.

Component	Value / Part	Reason
TVS Diode	SMBJ33A	Clamps voltage spikes above 33V — protects 25.2V 6S rail
Bulk Capacitor	1000 μ F 50V	Acts as an energy reservoir, supplies transient current during rapid motor load changes, and reduces battery voltage dips and switching ripple
Battery Connector	XT60	Low-resistance, rated for high current

3. 12V Buck Converter

Component: LMR33630BDDA

Schematic circuit:



Explanation:

The 6S LiPo battery can supply up to 25.2V and the FD6288Q gate driver needs a regulated 12V supply to properly drive the MOSFET gates. The battery voltage is reduced to 12V using an LMR33630 buck converter with low power loss, as opposed to a linear regulator.

Component	Value	Reason for selection
C3	10µF, 50V	Provides bulk input filtering and reduces battery voltage ripple during switching. The 50V rating provides sufficient safety margin above the maximum battery voltage of 25.2V.
C4	220nF X7R, 25V	High-frequency bypass capacitor placed close to the VIN pin to suppress switching noise. The X7R dielectric ensures stable capacitance over temperature and voltage variations.
C12	100nF, 50V	Bootstrap capacitor required by the internal high-side gate driver of the LMR33630. The datasheet recommends a 100nF value for reliable switching operation.
L1	6.8µH	Selected according to the switching frequency and output current requirements to maintain continuous current flow and minimize output current ripple.
C5	100nF, 50V	High-frequency output filter capacitor that suppresses switching noise and EMI

C6, C13	22μF, 25V each	Output capacitors that smooth the 12V output voltage and reduce ripple during load transients. Using two capacitors increases effective capacitance and lowers ESR, improving transient response and output stability.
R1	110kΩ	Upper feedback resistor selected to set the output voltage to approximately 12V.
R2	10kΩ	Lower feedback resistor used with R1 to form the feedback divider required for output voltage regulation.

Feedback resistor calculation:

The output voltage of the buck converter is calculated by feedback resistor network:

$$V_{OUT} = V_{REF} \left(1 + \frac{R_1}{R_2} \right)$$

$$V_{OUT} = 1 \left(1 + \frac{110k\Omega}{10k\Omega} \right) = 12V$$

Thus, R1 = 110kΩ and R2 = 10kΩ are selected to generate a regulated 12V output. In the hardware we choose R1 = one 10kΩ series with 100kΩ and R2 = 10kΩ.

Working Principle:

The LMR33630 buck converter works by turning the input voltage ON and OFF at a high frequency. In the ON state the energy is stored in the 6.8μH inductor and in the OFF state the energy is released to the output. The 22μF output capacitors smooth the switching waveform into a steady 12V DC supply. The feedback network formed by R1 and R2 continuously senses the output voltage and adjusts the switching duty cycle to maintain the output regulated at 12V against changing battery and load conditions.

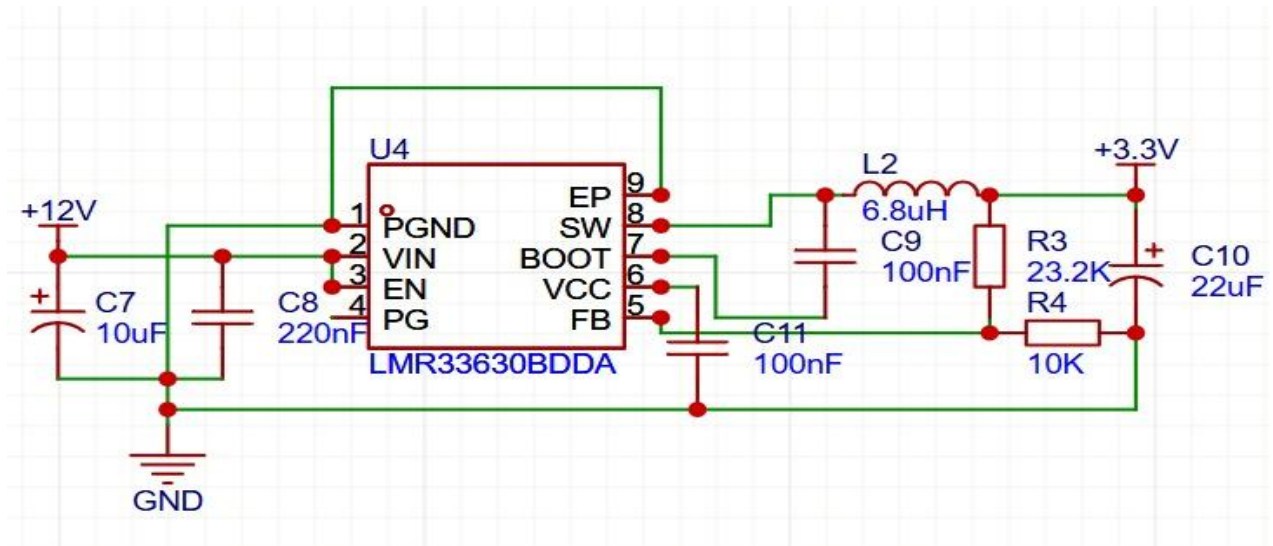
Why This Circuit Is Needed:

This circuit efficiently converts the 6S LiPo battery voltage (18V to 25.2V) to a regulated 12V supply for the FD6288Q gate driver. It offers high conversion efficiency with low heat generation, stable output under fast motor load changes, low voltage ripple and switching noise, and reliable operation of the gate driver and the overall ESC power electronics system.

4. 3.3V Buck Converter

Component: LMR33630BDDA (second stage)

Schematic circuit:



Explanation:

The first buck converter provides a regulated 12V supply. The 12V supply is then stepped down to 3.3V using a second LMR33630 buck converter. The 3.3V rail is needed to power the STM32G071 microcontroller, INA180 current-sense amplifiers and other sensing circuitry. This is much more efficient than the linear regulator and less heat is generated. This ensures a stable and reliable power supply for the ESC control electronics.

Component	Value	Reason for Selection
C7	10µF, 25V	Provides bulk input filtering for the 12V input rail and reduces input voltage ripple caused by converter switching.
C8	220nF X7R, 25V	High-frequency bypass capacitor placed close to the VIN pin to suppress switching noise and improve converter stability. The X7R dielectric provides stable capacitance over temperature and voltage variations.
C11	100nF, 50V	Bootstrap capacitor required by the internal high-side gate driver of the LMR33630. The datasheet recommends a 100nF value for reliable operation.
L2	6.8µH	Selected according to the switching frequency and output current requirements to maintain continuous current flow and minimize output current ripple.

C9	100nF, 50V	High-frequency output filter capacitor used to suppress switching noise and reduce electromagnetic interference (EMI).
C10	22μF, 25V	Output capacitor that smooths the regulated 3.3V output and reduces voltage ripple during transient load conditions.
R3	23.2kΩ	Upper feedback resistor selected to configure the output voltage to approximately 3.3V.
R4	10kΩ	Lower feedback resistor used with R3 to form the feedback divider required for output voltage regulation.

Feedback resistor calculation:

The output voltage of the buck converter is calculated by feedback resistor network:

$$V_{OUT} = V_{REF} \left(1 + \frac{R_3}{R_4} \right)$$

$$V_{OUT} = 1 \left(1 + \frac{23.2k\Omega}{10k\Omega} \right) = 3.32V \text{ Approx. } = 3.3V$$

Working Principle:

The LMR33630 is a buck converter. It “switches” the 12V ON and OFF at a very high frequency. In the ON state the 6.8uH inductor (L2) stores energy and in the OFF state the stored energy is delivered to the output. The output capacitor is 22uF (C10), filtering the switching waveform to a steady 3.3V DC power supply. The output voltage is constantly monitored by the feedback network of R3 and R4 and the switching duty cycle is adjusted so that the output is regulated at 3.3V, regardless of load conditions.

Why This Circuit is Needed:

The circuit efficiently converts the regulated 12V supply to the 3.3V required by the STM32G071 microcontroller, INA180 current sense amplifiers and sensing circuits. The ESC control electronics work reliably, and the motor is controlled precisely with high efficiency, low heat generation, and a stable output voltage.

PA7	CS_C	Phase C Current Sense	ADC_IN7
PB6	BEMF_C	Phase C Back-EMF Detection (MCP6561 Comparator Output)	GPIO / EXTI
PB7	VREF	Virtual Neutral Reference for Comparator	GPIO / ADC
PA8	HIN1	High-Side PWM Output – Phase A	TIM1_CH1
PA11	HIN2	High-Side PWM Output – Phase B	TIM1_CH2
PA12	HIN3	High-Side PWM Output – Phase C	TIM1_CH3
PB3	LIN1	Low-Side PWM Output – Phase A	GPIO / Timer Output
PB4	LIN2	Low-Side PWM Output – Phase B	GPIO / Timer Output
PB5	LIN3	Low-Side PWM Output – Phase C	GPIO / Timer Output
PA13	SWDIO	Serial Wire Debug Data	SWD
PA14	SWCLK	Serial Wire Debug Clock	SWD
PF2	NRST	External Reset Input	Reset
VDD/VDDA	+3.3V	MCU Power Supply	Power
VSS/VSSA	GND	Ground Reference	Ground

Working Principle:

The sensor less BLDC ESC is controlled by the STM32G071GBU6TR as the main controller. It constantly monitors the three current sensing channels (CS_A, CS_B, and CS_C) and the three Back-EMF sensing channels (BEMF_A, BEMF_B, and BEMF_C) to determine the motor's operational state and rotor position. From this information, six PWM control signals (HIN1-HIN3 and LIN1-LIN3) are generated by the microcontroller and sent to the gate driver FD6288Q. The gate driver drives the six MOSFETs of the three-phase inverter to power the BLDC motor.

The ESC uses Back-EMF zero-crossing detection to detect the rotor position, which is used for sensor less commutation. A virtual neutral reference voltage (VREF) is formed by summing the voltages of the three motor phases through a resistor network:

$$V_{REF} = \frac{V_{PhaseA} + V_{PhaseB} + V_{PhaseC}}{3}$$

The rotor position is determined by comparing the Back-EMF signal of each floating phase with this virtual neutral reference. The hardware zero crossing detection uses the internal analog comparators of the STM32 for phase A (BEMF_A) and phase B (BEMF_B). For Phase C (BEMF_C) a digital zero-crossing signal is output to the STM32 by an external MCP6561 comparator that compares the phase voltage with VREF. The controller detects these zero crossings to determine the correct

commutation instant and advance the inverter switching sequence without the need for Hall-effect sensors.

We monitor current today with low-value shunt resistors and current sense amplifiers. The amplified current signals (CS_A, CS_B and CS_C) are passed to the STM32 ADC channels. The microcontroller continuously samples these currents to monitor motor load, calculate operating conditions, and detect abnormal situations such as overcurrent or stall conditions. This enables protection features and provides valuable feedback for motor control.

PWM generation with the advanced timer interfaces of the STM32. The microcontroller generates six PWM signals (three high-side outputs: HIN1-HIN3 and three low-side outputs: LIN1-LIN3) based on the desired motor speed and commutation state. These PWM signals are used to control the duty cycle applied to the motor phases, controlling motor speed and torque. The PWM outputs are driven to the FD6288Q gate driver that amplifies them to drive the MOSFET gates. It offers integrated 200 ns dead-time insertion and shoot-through protection, which allows safe inverter switching and improves the overall system reliability.

Why this circuit is Needed:

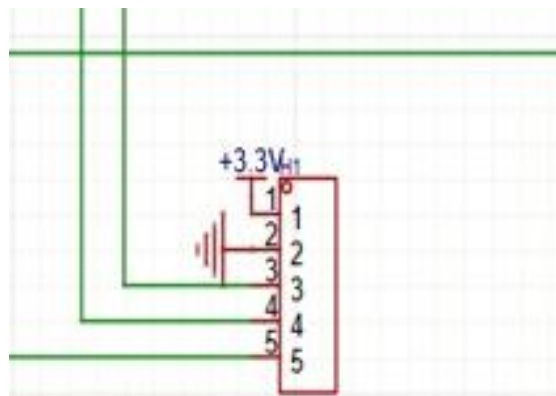
This circuit is designed to be the main control unit of sensor less BLDC ESC, used for motor control, rotor position estimation and system protections. The STM32G071 has internal comparators for BEMF_A and BEMF_B, and an external MCP6561 comparator for BEMF_C to process the Back-EMF signals to determine the rotor position without Hall sensors. The current-sensing circuits continuously monitor the three phase currents for overcurrent protection and performance monitoring. The sensed information is fed into the microcontroller, which generates the PWM signals to control the FD6288Q gate driver and the three-phase MOSFET inverter. The virtual neutral reference (VREF) allows accurate Back-EMF zero-crossing detection for accurate sensor less commutation. The combination of these functions enables efficient motor operation, reliable fault protection and safe operation of the drone propulsion system.

6. SWD Header Interface

Pins used

- PA13 → SWDIO (Serial Wire Debug Data)
- PA14 → SWCLK (Serial Wire Debug Clock)
- PF2 → NRST (Reset pin)

Schematic circuit:



1x5 SWD Header

Working Principle:

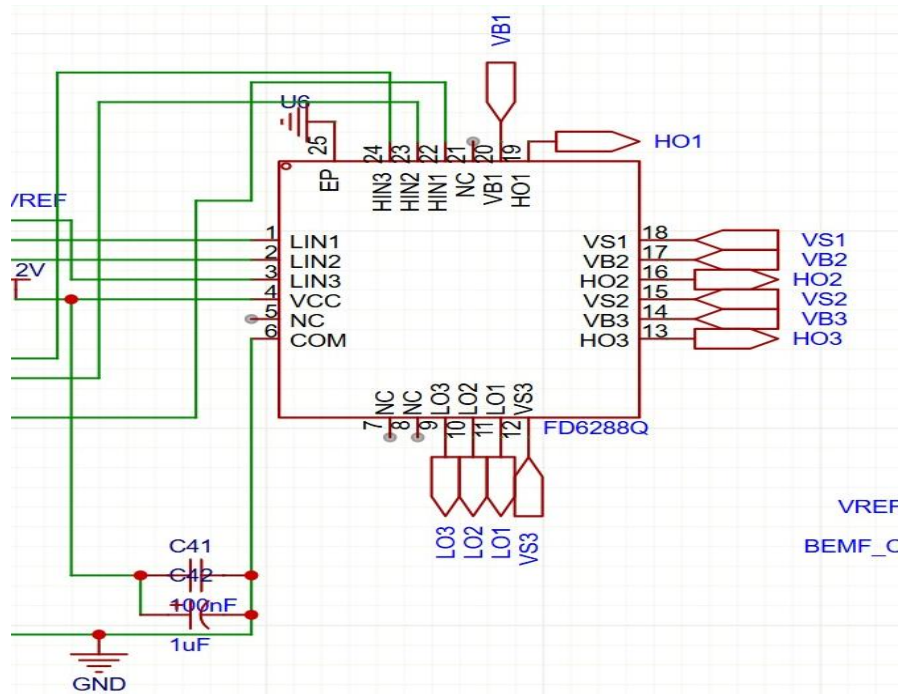
The SWD header is a direct communication interface between the STM32G071 microcontroller and an external programmer/debugger such as an ST-Link. The SWDIO, SWCLK and NRST signals enable the firmware to be downloaded to the microcontroller and the system to be debugged during the development and test phase.

Why This Circuit Is Needed:

The SWD header is for programming, debugging and validation of the ESC firmware during development. It can be programmed directly from a computer with an ST-Link programmer, allowing trouble shooting of hardware and software problems before deployment in the drone system.

7. FD6288Q Gate Driver

Schematic circuit:



Working Principle:

The FD6288Q is an interface between an STM32G071 microcontroller and a MOSFET inverter stage. The STM32 generates low power 3.3V PWM control signals (HIN1–HIN3 and LIN1–LIN3) to the inputs of the FD6288Q. These signals are received by the gate driver, which amplifies them to the proper gate-drive voltage and current to rapidly switch the six IRFB7545PBF MOSFETs. The FD6288Q has bootstrap circuitry to drive the high-side and low-side MOSFETs required for three-phase BLDC motor operation. It also has built in dead time and shoot through protection to prevent MOSFET pairs from conducting at the same time.

Why This Circuit Is Needed:

The STM32G071 microcontroller can only generate 3.3V low current PWM signals which are not strong enough to drive the gates of the six power MOSFETs directly. This is the reason of the need of the FD6288Q gate driver. The FD6288Q amplifies these signals and provides the high-current gate drive required for efficient MOSFET switching. It also supports high side gate driving with bootstrap circuitry, enabling operation of the three-phase inverter. In addition, the FD6288Q features an integrated 200ns dead-time insertion and shoot-through protection, which helps to

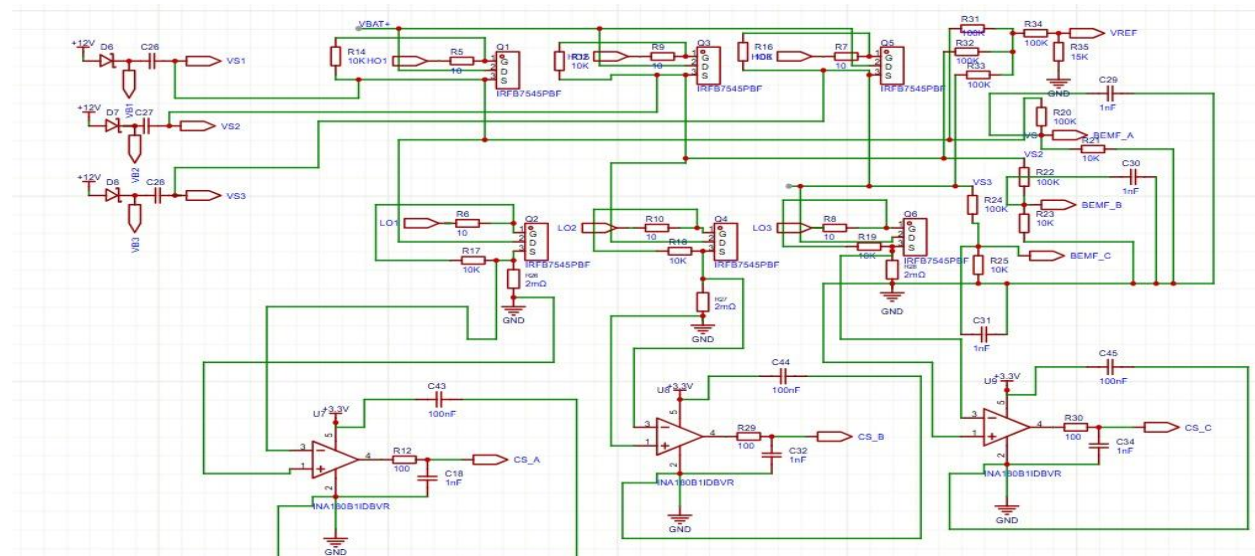
avoid simultaneous conduction of high-side and low-side MOSFETs, thereby increasing system reliability and protecting the inverter from damage. This leads to safer operation and lower switching losses and more efficient motor control for the drone ESC.

Key Features of FD6288Q Relevant to This Design:

Feature	Value	Benefit
Built-in dead-time	200 ns (typ.)	Reduces risk of shoot-through by inserting delay between high-side and low-side switching.
Shoot-through protection	Yes	Prevents simultaneous conduction of high-side and low-side MOSFETs in the same phase leg.
Input logic level	3.3V / 5V compatible	Direct interface with STM32 GPIOs without level shifters.
VCC supply range	5V – 20V	Allows operation from the regulated 12V gate-driver supply.
Input signal filtering	Built-in	Improves noise immunity against switching transients and EMI.
Bootstrap support	Yes	Supports floating high-side gate drive using VBx and VSx pins.
Peak gate drive current	±1.5A (typ.)	Enables fast charging and discharging of MOSFET gate capacitances, reducing switching losses.

8. Three Phase Inverter Circuit

Schematic circuit:



Components Used:

- IRFB7545PBF N-Channel MOSFET × 6
- 10Ω Gate Resistors × 6
- 10kΩ Gate Pull-Down Resistors × 6

Purpose:

The main power conversion stage of the ESC is the three-phase inverter. Its main job is to convert the DC voltage that comes from the 6S LiPo battery into a controlled three-phase output that is required to drive the BLDC motor. The inverter itself consists of six IRFB7545PBF MOSFETs arranged as a three-phase bridge with each phase of the motor connected between a high-side and low-side MOSFET. The ESC switches these mosfets in the right order to create a rotating magnetic field inside the motor stator. This makes the rotor rotate.

The control signals for the PWM generation are provided by the STM32G071 microcontroller based on the desired motor speed and the rotor position information received from the Back-EMF sensing circuit. These signals are amplified by the FD6288Q gate driver and applied to the MOSFET gates. During six-step commutation, two motor phases are energized at a time while the third phase remains floating. The conducting phases are continuously changed according to the rotor position, producing smooth and efficient motor operation.

Why IRFB7545PBF Was Selected:

The IRFB7545PBF was chosen because it satisfies the voltage, current, efficiency, and switching requirements of a high-performance drone ESC. The 60V drain-to-source voltage rating gives a good safety margin above the maximum 25.2V battery voltage and protects against switching transients. The 110A continuous drain current rating means it can handle high motor currents and startup surges, while its low RDS(ON) of 4.7mΩ minimizes conduction losses, improves efficiency and reduces heat generation. The MOSFET is compatible with the FD6288Q's 10-12V gate drive and can efficiently operate at switching frequencies up to 32kHz. A 10Ω gate resistor is used to limit the gate current, reduce ringing, and suppress EMI. A 10kΩ gate-to-source pull-down resistor is used to keep the MOSFET in the OFF state during startup or fault conditions.

Key Specifications of the IRFB7545PBF MOSFET:

Parameter	IRFB7545PBF	Requirement	Benefit
Breakdown Voltage (VDS)	60V	Must exceed 25.2V battery voltage and switching spikes	Provides sufficient voltage margin and protects against transient overvoltage conditions.
Continuous Drain Current	110A	Must handle high motor current	Supports high startup currents and peak motor loads without overheating.
RDS(ON)	4.7mΩ	Low conduction losses required	Reduces power loss, improves efficiency, and minimizes heat generation.
Gate Threshold Voltage (VGS(th))	4V max	Compatible with 10–12V gate drive	Ensures reliable switching with the FD6288Q gate driver.
Package	TO-220	Good thermal performance	Provides efficient heat dissipation and simplifies mounting and prototyping.

Why this 3 phase Inverter circuit is needed:

The three-phase MOSFET inverter is essential because a BLDC motor cannot operate directly from the DC output of a battery. The inverter converts the battery's DC voltage into a controlled three-phase waveform required for motor operation. The MOSFETs provide efficient high-current switching, while the gate resistors and pull-down resistors ensure reliable and safe switching behaviour. Together with the gate driver and control circuit, the inverter enables precise motor speed control, high efficiency, reliable operation, and safe power delivery for the drone propulsion system.

Complete Component Summary Table:

Component	Value / Part Number	Purpose
Q1–Q6	IRFB7545PBF	Six N-channel MOSFETs forming the three-phase inverter bridge to drive the BLDC motor.
R5–R10	10Ω	Gate resistors that limit gate charging/discharging current, reduce ringing, and suppress EMI during MOSFET switching.
R14–R19	10kΩ	Gate-to-source pull-down resistors that keep MOSFETs OFF during startup, reset, and fault conditions.
Phase A, B, C	Motor Phase Outputs	Three inverter outputs connected to the BLDC motor windings.
R26–R28	2mΩ	Precision shunt resistors used to measure phase current by generating a small voltage proportional to motor current.
R20, R22, R24	100kΩ	Upper resistors of the Back-EMF voltage divider network that reduce motor phase voltage to a safe level for sensing.
R21, R23, R25	10kΩ	Lower resistors of the Back-EMF voltage divider network used with the 100kΩ resistors to scale the phase voltages for MCU and comparator inputs.
C29, C30, C31	1nF	Noise-filter capacitors that remove high-frequency switching noise from the Back-EMF signals, improving zero-crossing detection accuracy.

R31, R32, R33	100kΩ	Resistor network used to generate the virtual neutral reference voltage (VREF) from the three motor phase voltages.
R34	100kΩ	Series resistor between the virtual neutral network and the VREF output. It isolates the VREF node, limits transient currents, and improves stability of the reference signal.
R35	15kΩ	Pull-down resistor used in the VREF network to stabilize the virtual neutral voltage.
BEMF_A	Phase A Back-EMF Signal	Used by the STM32 internal comparator for rotor position detection.
BEMF_B	Phase B Back-EMF Signal	Used by the STM32 internal comparator for rotor position detection.
BEMF_C	Phase C Back-EMF Signal	Connected to the external MCP6561 comparator for zero-crossing detection.
VREF	Virtual Neutral Reference	Reference voltage calculated from the average of the three phase voltages and used for Back-EMF comparison.

Voltage divider calculation:

The Back-EMF voltage is scaled according to the voltage divider equation:

$$V_{BEMF} = V_{PHASE} \times \frac{R_{LOWER}}{R_{UPPER} + R_{LOWER}}$$

$$V_{BEMF} = V_{PHASE} \times \frac{10k\Omega}{100k\Omega + 10k\Omega}$$

$$V_{BEMF} = V_{PHASE} \times 0.0909$$

$$V_{BEMF} = 25.2 \times 0.0909$$

$$V_{BEMF} = 2.29V$$

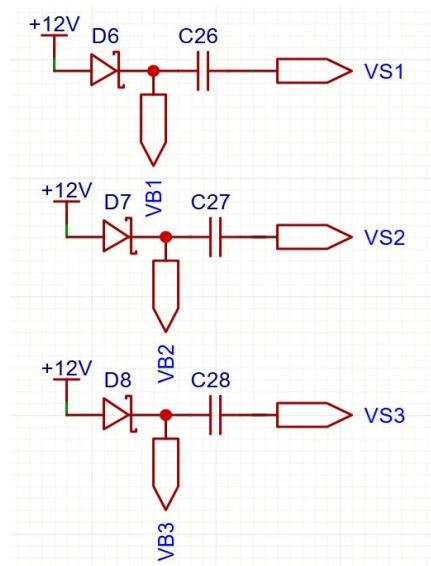
Conclusion: The 100kΩ / 10kΩ divider scales the max 25.2V motor phase voltage to approx. 2.29V, which is safely within the STM32's 3.3V input range for Back-EMF sensing.

Signal Summary Table:

Signal	Destination	Purpose
HO1, HO2, HO3	High-Side MOSFET Gates	Drive the high-side MOSFETs of phases A, B, and C.
LO1, LO2, LO3	Low-Side MOSFET Gates	Drive the low-side MOSFETs of phases A, B, and C.
BEMF_A	STM32 Comparator	Phase A zero-crossing detection.
BEMF_B	STM32 Comparator	Phase B zero-crossing detection.
BEMF_C	MCP6561 Comparator	Phase C zero-crossing detection.
VREF	STM32 / Comparator Reference	Virtual neutral reference for sensor less commutation.
Phase A, B, C	BLDC Motor	Three-phase motor outputs.

9. Boot Strap Circuit

Schematic circuit:



Components Used:

Component	Value / Part Number	Purpose
D6, D7, D8	SS54 Schottky Diode	Charges the bootstrap capacitor from the 12V supply and prevents reverse current flow.
C26, C27, C28	220nF X7R, 50V	Bootstrap capacitors that store energy required to drive the high-side MOSFET gates.
VB1, VB2, VB3	FD6288Q Bootstrap Supply Pins	Positive floating supply terminals for the high-side gate drivers.
VS1, VS2, VS3	FD6288Q Switching Node Pins	Connected to the motor phase nodes and act as the reference point for the high-side gate driver.

Why SS54 Schottky Diode is chosen:

The bootstrap circuit chosen the SS54 Schottky diode as it has a low forward voltage drop of 0.55V, allowing the bootstrap capacitor to charge to nearly the full 12V supply voltage. This allows for enough gate drive voltage for the high-side MOSFETs. It is a Schottky diode with very fast switching characteristics and virtually no reverse recovery time, which makes it suitable for high frequency ESC operation up to 32kHz. The diode is rated for 5A which is plenty for the bootstrap charging current and 40V is more than enough for the reverse voltage rating for a 6S ESC system. Plus, its SMA surface-mount package is small, lightweight and perfect for PCB-based drone ESC designs.

How Bootstrap Circuit Works in Our Esc Design:

In our ESC the gate driver FD6288Q has to drive the low side MOSFETs (Q2, Q4, Q6) and the high side MOSFETs (Q1, Q3, Q5). Driving the low-side MOSFETs is simple because their source terminals are connected close to GND. The source terminals of the high-side mosfets are however connected to the motor phase nodes (VS1, VS2, VS3), which are constantly oscillating between 0V and battery voltage (up to 25.2V).

For example consider Phase A:

- **D6** = SS54 bootstrap diode
- **C26** = 220nF bootstrap capacitor
- **VB1** = Bootstrap supply pin
- **VS1** = Phase-A switching node

Step1: Boot strap capacitor charging

When the low-side MOSFET (Q2) turns ON:

- VS1 is pulled close to 0V (GND).
- Current flows from:

$$12V \rightarrow D6 \rightarrow C26 \rightarrow VS1 \rightarrow GND$$

- The capacitor charges to approx.:

$$V_{BOOT} = 12V - 0.55V$$
$$V_{BOOT} = 11.45V$$

Where 0.55V is the forward voltage drop of the SS54 diode.

At this point:

$$VB1 - VS1 = 11.45V$$

Step2: High Side MOSFET Turn On

Now the FD6288Q gate driver receives a command to turn ON Q1 (high-side MOSFET).

Suppose the motor phase voltage rises to:

$$VS1 = 25.2V$$

The charged bootstrap capacitor "floats" upward with VS1 and maintains:

$$VB1 = VS1 + 11.45V = 25.2V + 11.45V = 36.65V$$
$$V_{GS} = VB1 - VS1 = 36.65V - 25.2V = 11.45V$$

This 11.45V gate-to-source voltage fully turns ON the high-side MOSFET.

Step3: Motor Continues Running

The six-step commutation proceeds as:

- Q2 periodically turns ON again.
- C26 recharges through D6.
- The same process happens for:
 - D7 + C27 → VB2/VS2 (Phase B)
 - D8 + C28 → VB3/VS3 (Phase C)

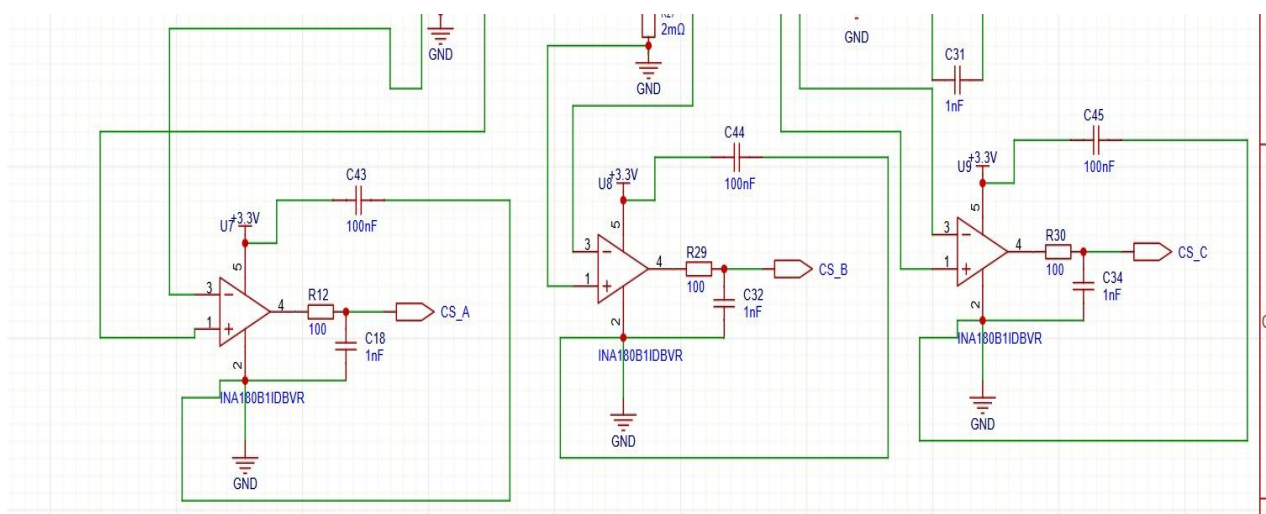
Thus, each phase gets its own floating high side supply.

Why Bootstrap Circuit is Needed:

The bootstrap circuit is crucial because the ESC uses N channel MOSFETs on the high side of the 3-phase inverter. To fully turn ON a high-side MOSFET, the gate voltage needs to be about 10-12V higher than the source voltage. The source terminal of the high side MOSFET is connected to the motor phase node and can go up to the battery voltage (25.2V for a 6S system). A fixed 12V supply cannot provide a gate voltage higher than this phase voltage. The bootstrap circuit solves the problem by using a charged bootstrap capacitor to provide a floating supply voltage referenced to the motor phase node. So, the FD6288Q can give us the gate-source voltage around 11.45V and the high-side MOSFETs can be turned ON fully. Without the bootstrap circuit, the high-side MOSFETs could not be driven properly, preventing the three-phase inverter from operating and making BLDC motor control impossible.

10. Current Sensing Circuit:

Schematic circuit:



Components used:

Component	Value / Part Number	Purpose
R26, R27, R28	2mΩ Shunt Resistors	Generate a small voltage proportional to the current flowing in each motor phase.
U7, U8, U9	INA180B1IDBVR	Amplify the small shunt voltage for STM32 ADC measurement.
R12, R29, R30	100Ω	Output protection resistors that limit transient currents and help reduce switching noise at the ADC input.
C18, C32, C34	1nF	Noise filter capacitors that remove high-frequency switching noise from the current-sense signals.
C43, C44, C45	100nF	Decoupling capacitors that stabilize the INA180 power supply and reduce noise.
CS_A, CS_B, CS_C	Analog Outputs	Current-sense signals connected to STM32 ADC inputs for current monitoring and protection.

Why We Use Three Current Sensing Circuits:

In a 3-phase BLDC motor, the current in each phase is not always the same during commutation. The separate current sensing circuits are used for Phase A, Phase B and Phase C.

Using three current sensors allows the STM32 to:

- Check current in each motor phase separately.
- Detect phase imbalance or abnormal condition.
- Provide accurate overcurrent protection.
- Improve the performance of motor control.
- Keep an eye on motor load and power consumption.

Without current sensing the ESC would have no idea how much current the motor is pulling. This can lead to MOSFET, battery or motor damage.

Working Principle:

Each motor phase contains a 2mΩ shunt resistor placed in the current path. When motor current flows through the shunt resistor, a very small voltage is generated according to the ohm's law:

$$V=IR$$

For example:

$$V_{SHUNT} = 30A \times 2m\Omega = 60mV$$

This voltage is too small to be measured directly by the STM32 ADC, so it is amplified by the INA180B1 current-sense amplifier.

The INA180B1 has a fixed gain of approximately 26 V/V therefore:

$$V_{OUT}=26 \times V_{SHUNT}$$

For a 30A motor current:

$$V_{OUT} = 26 \times 0.06V = 1.56V$$

The outputs CS_A, CS_B and CS_C are amplified and sent to the STM32 ADC pins for the microcontroller to continuously monitor the motor current and do protection functions.

Current Sensing Calculation:

Motor Current (A)	Shunt Voltage (mV)	CS Output Voltage (V)
0 A	0 mV	0.00 V
5 A	10 mV	0.26 V
10 A	20 mV	0.52 V
30 A	60 mV	1.56 V
60 A	120 mV	3.12 V

The output voltage at 60A is around 3.12V, which is well inside the 3.3V ADC input range of the STM32 although uses almost all of the ADC measurement range to give better accuracy. For currents above 60A, the output voltage tends to approach and then exceed the 3.3V ADC limit, resulting in ADC saturation and a decrease in measurement accuracy. In these conditions the ESC firmware can read the saturated value as an over current condition and take protective measures to avoid damage to the MOSFETS, motor and battery.

Why This Current Sensing Circuit Is Needed:

The three-phase current sensing circuit is used to measure the flow of current through each motor phase (Phase A, Phase B and Phase C) of the BLDC motor in real time. The amplified current signals (CS_A, CS_B and CS_C) are observed with the STM32G071 ADC to monitor the motor load conditions and phase currents during the commutation cycle. This information is used to implement overcurrent protection, protection from damage of the MOSFETS, battery and motor and to improve the overall reliability of the ESC. Separate measurement of each of the three phases also aids in the detection of abnormal operating conditions i.e., phase imbalance, excessive current draw or motor faults. This allows the ESC to work more safely, efficiently and reliably in various drone flight scenarios.

Overall System Operation

Power Flow:

- The LiPo battery is a 6S (18V – 25.2V) that powers the ESC through the input connector XT60.
- The SMBJ33A TVS diode, 1000μF bulk capacitor, and 100nF ceramic capacitor protect the ESC from voltage spikes, battery transients, and switching noise.
- The first LMR33630 buck converter reduces the battery voltage to a regulated 12V supply.
- The 12V rail powers the FD6288Q gate driver and the charge source for the bootstrap circuits.
- Another LMR33630 buck converter takes the 12V and regulates it down to 3.3V.
- The STM32G071 microcontroller, INA180 current-sense amplifiers, MCP6561 comparator and associated sensing circuits are powered from the 3.3V rail.

Control Flow – Sensor less Six Step Commutation

- Once powered up, the STM32G071 sets up the PWM timers, ADC channels, comparators and protection routines.
- The STM32 generates six PWM control signals (HIN1~HIN3 and LIN1~LIN3) according to the throttle command and motor working condition.
- The PWM signals are fed into the FD6288Q gate driver, which adds in its own built-in dead-time of 200ns, and drives the gates of the MOSFETs with the regulated 12V supply.
- The bootstrap circuits (SS54 diodes + 220nF capacitors) supply the floating gate-drive voltages for the high-side MOSFETs.
- The six IRFB7545PBF MOSFETs apply the battery voltage to the motor windings according to the six-step commutation sequence.
- At any given moment, one phase is connected to VBAT with a high-side MOSFET, one phase is connected to ground with a low-side MOSFET and the third phase is floating.
- The floating phase produces a BEMF voltage that includes rotor position information.
- The resistor-divider networks scale the motor phase voltages to safe levels for the STM32 and comparator inputs.
- A virtual neutral reference voltage (VREF) is formed by summing the voltages of the three motor phases through a resistor network:
 - $$V_{REF} = \frac{V_{PhaseA} + V_{PhaseB} + V_{PhaseC}}{3}$$
- The STM32 internal comparators detect the zero crossings of BEMF_A and BEMF_B with VREF as reference.
- The external comparator MCP6561 is used to detect the BEMF_C zero crossings and send the result to STM32.

- When a zero crossing is detected the STM32 estimates the rotor position and applies the required commutation delay before switching to the next commutation state.
- This cycle repeats continuously. This enables sensor less operation of BLDC motors without Hall-effect sensors.

Current Monitoring and Protection:

- Three 2mΩ shunt resistors measure the current flowing through motor phases A, B and C.
- The shunt voltages are amplified by the INA180B1 amplifiers to produce the signals CS_A, CS_B and CS_C.
- The STM32 is continuously sampling those signals on its ADC channels to sense the motor current.
- If the measured current exceeds the programmed safety threshold the firmware can reduce the PWM duty cycle or even turn off the inverter to protect the ESC.
- The current sensing system also provides motor load and performance monitoring.